



A quick paster type of soluble nanoparticle microneedle patch for the treatment of obesity

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ABSTRACT

Obesity is a major public burden on the working population and induces chronic diseases. Its treatment often requires long-term medication, which makes patient compliance difficult. In this study, we reported the value of HORN-MN, which comprised a fast-soluble hyaluronic acid microneedle matrix and a weak acid-degradable oleanolic acid dimer of rosiglitazone nanoparticles. The results showed that the microneedles easily punctured the stratum corneum and dissolved in the dermis of the abdominal wall within 5 min, followed by the release of rosiglitazone nanoparticles. Thereafter, the nanoparticles were endocytosed by macrophages and white adipocytes, then degraded to oleanolic acid in the lysosomes, thereby, releasing rosiglitazone. Oleanolic acid significantly improved the inflammatory status of obese adipose tissue and promoted white adipocyte browning, and rosiglitazone significantly potentiated WAC browning. Accordingly, the patch demonstrated a remarkable obesity-reducing efficacy in mice. In conclusion, this study developed a quick paster type of soluble rosiglitazone nanoparticle microneedle for the treatment of obesity. This patch can be suitable for working people, with an evident obesity-reducing efficacy but no effect on skin integrity despite multiple administrations.

1. Introduction

Obesity is a major public health problem, with more than 2 billion adults worldwide being overweight and obese [1]. Severe subcutaneous fat accumulation is the main feature in patients who are obese. Evidence has shown that fat accumulation is closely related with the pathological changes in the subcutaneous adipose tissue and that the excessive adipocytes recruit macrophages to form an inflammatory microenvironment [2], in which the presence of abundant inflammatory factors continuously promote the enlargement of white adipocytes (WAC), leading to obesity [3]. Studies have shown that inducing WAC to convert to beige adipocytes (BAC) can consume excessive energy and, therefore, has the potential to treat obesity [4]. However, induction of adipocyte browning is often hampered by an inflammatory environment. Consequently, improvement of the inflammatory microenvironment in

subcutaneous adipose tissue and effective induction of adipocyte browning are the critical scientific issues to be addressed.

Here, we reported on the use of a quick paster type of soluble rosiglitazone nanoparticle microneedle for improving the inflammatory microenvironment in subcutaneous adipose tissue and inducing adipocyte browning. The patch contained hyaluronic acid (HA), which was used as the matrix of the soluble microneedle because of its ideal biocompatibility and mechanical strength, fast dissolution rate in interstitial fluid, and capability to retain skin moisture [5]. Moreover, oleanolic acid (OA) was selected as the agent to correct adipose inflammation, and rosiglitazone (Rosi) was the agent used to induce adipocyte transformation. In addition, oleanolic acid dimer (HOA) was newly synthesized as the nanoparticle material for preparing the patch, which we called soluble rosiglitazone nanoparticle microneedle patch (HORN-MN).

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Subcutaneous adipose tissue mainly comprises adipocytes, adipose precursor cells, immune cell-like macrophages, blood vessels, and extracellular matrix [6]. In obesity, subcutaneous WAC contain large single-locular lipid droplets that store excessive energy in the form of triglycerides (TG). For treating obesity, WACs can be changed to beige adipocytes by peroxisome proliferator-activated receptor γ (PPAR γ) agonists, such as rosiglitazone [7]. Beige adipocytes can induce weight loss, because they contain abundant mitochondria and have relatively strong energy metabolism; highly express uncoupling protein 1 (UCP1), which contributes to the inhibition of mitochondrial oxidative phosphorylation; and promote free fatty acid decomposition [4].

In a state of inflammation, subcutaneous WAC contain a large amount of TG, which leads to adipocyte hypoxia and endoplasmic reticulum stress, which result in the recruitment of M1 macrophages and secretion of tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), interleukin-6 (IL-6), and other pro-inflammatory factors [3]. These proinflammatory factors further hinder the transforming process to beige adipocytes. In particular, TNF α lowers UCP1 expression, thereby, hindering the uncoupling process during mitochondrial oxidative phosphorylation and increasing fat accumulation [8].

Oleanolic acid, a pentacyclic triterpenoid that can be isolated from the fruit of chasteberry and is clinically used for the treatment of inflammatory disorders, such as acute jaundice hepatitis, chronic viral hepatitis, and gastric ulcer [9]. Its mechanism is associated with the downregulation of proinflammatory factors through inhibition of the nuclear factor kappa-B (NF- κ B) signaling pathway [10,11]. Because of its unique chemical structure, OA can be synthesized into a dimer material, which can self-assemble into the hydrazone bond-linked HOA matrix nanoparticle (HOAN). This nanoparticle has been used to encapsulate rosiglitazone and can be degraded into the original OA in cells to exert the simultaneous therapeutic effects of OA and rosiglitazone.

Microneedle patch is a promising transdermal drug delivery system that can overcome the skin barrier for drug penetration with minimal skin irritation or injury [12]. A microneedle array ranging from 250 to 1000 μ m in length on the patch can painlessly puncture the stratum corneum and deliver the drug directly to the subcutaneous tissue [13]. The primary manifestation of obesity is accumulation of subcutaneous fat in the waist and abdomen. Inducing subcutaneous WAC browning is a promising novel approach to combat obesity. Currently, there is an exciting treatment strategy of utilizing microneedles to directly deliver adipose-inducing drugs, such as rosiglitazone, to subcutaneous adipose tissue for weight loss [14,15]. However, the impact of adipose tissue inflammation on WAC browning remains inadequately explored [16], and formulating strategies to modulate the inflammatory milieu in adipose tissue to enhance WAC browning poses a significant challenge in obesity treatment.

The objective of the present study was to develop a quick paste type of soluble rosiglitazone nanoparticle microneedle, which enables HOA to self-assemble into HOAN. We verified the efficacy and relevant mechanism of this patch in improving subcutaneous inflammation and inducing beige adipocyte transformation. The study was performed through rational design, construction and structural characterization, mechanism analysis, and functional verification in vitro and in obese mice.

2. Materials and methods

2.1. Materials

Rosi and OA were purchased from Shanghai Macklin Biochemical (Shanghai, China). Negative microneedle mold was purchased from Micro Core Medicine Technology (Taizhou, Jiangsu, China). 1,2-Distearoyl-*sn*-glycero-3-phosphatidylethanolamine-N-[methoxy(polyethylene glycol)-2000] (DSPE-PEG₂₀₀₀) was purchased from Avanti Polar Lipid (Alabaster, AL, USA). HA was from Rhawn (10k Da; Beijing

local agent, China).

2.2. Animals

C57BL/6 mice (20 g, male) were purchased from Health Science Center of Peking University and maintained in a specific pathogen-free environment. Mice were maintained in the Peking University animal facility at 23 ± 2 °C with a 12 h light and dark cycle and had free access to food and water. All animal experiments were performed with the approval of the Committee for Experimental Animal Welfare of Biomedical Ethics of Peking University.

2.3. Synthesis and identification of nanomaterial

For synthesizing nanomaterial, an active intermediate of 2-(2-hydroxyethyl) hydrazone bond methanol (OH-Hz-OH) was firstly synthesized according to nucleophilic addition, with a little modification [17]. 2-(2-Aminoethylamino) ethanol and glycolaldehyde at a molecular ratio of 1:1 were dissolved in 95 % ethanol. The solution was concentrated by rotary evaporation at 40 °C. Then, absolute ethanol and anhydrous magnesium sulfate were added to the solution to remove water. After incubated at room temperature overnight, the reaction system was filtered to obtain the product of OH-Hz-OH. Afterward, OH-Hz-OH was measured by ¹H NMR, ¹³C NMR (AMX400, Bruker, Billerica, MA, USA), and infrared spectroscopy (Nexus 470, Thermo Fisher Scientific, Waltham, MA, USA), respectively. The yield of OH-Hz-OH was 83 %.

HOA was then synthesized through the esterification reaction between OH-Hz-OH and OA. OA (0.25 mmol), EDCI (0.30 mmol), and DMAP (0.30 mmol) were dissolved in anhydrous N, N-dimethylformamide (DMF, 4 mL), followed by adding N, N-diisopropylethylamine (DIPEA) to adjust pH to 7.0. OH-Hz-OH (0.13 mmol) was dropped into the reaction solution and then incubated at 45 °C for 6 h. The reaction mixture was washed three times with cold deionized water. The residue was collected and ultimately lyophilized to yield white powders of HOA. Afterward, HOA was measured by ¹H NMR spectroscopy and infrared spectroscopy, respectively. The yield of HOA was 86 %.

2.4. Preparation and characterization of the nanoparticles

For preparing HORN, a self-assembly nanoprecipitation approach was used [18]. HOA (20.0 μ mol) and rosiglitazone (11.2 μ mol) were dissolved in the combined reagent (tetrahydrofuran, acetone, and absolute ethanol = 4:3:3, v/v; 0.5 mL), and the above reagent was dropwise added into DSPE-PEG₂₀₀₀ water solution (1 mL, 3.5 mmol/L) under magnetic stirring to produce nanoparticle suspension. The suspension was then dialyzed against water (cut off 500 Da) for 30 min, thus producing the HORN. HOAN was the same prepared, except for adding rosiglitazone. Besides, the ester bond linked oleanolic acid trimer matrix rosiglitazone nanoparticle (ORN) was the same prepared by replacing POA with HOA.

The dynamic particle size, polydispersion index, and zeta potential were measured by ZetaSizer analysis (Nano ZS90, Malvern Instruments Ltd, Malvern, UK). The appearance of the nanoparticle was further observed by transmission electron microscopy (TEM, JEM1400Plus, JEOL, Tokyo, Japan).

2.5. Engineering of soluble rosiglitazone nanoparticle microneedle patch

HORN was homogeneously mixed with 30 % hyaluronic acid (300 mg/mL), added to the microneedle patch mold, and dried under vacuum for 10 min to form the tips of microneedles. Furthermore, 30 % hyaluronic acid was added onto the surface of the mold to form the backing layer, dried under vacuum for 30 min, and then desiccated at room temperature for 12 h, thus obtaining HORN-MN.

The control patches (RN-MN and HOAN-MN) were the same

prepared, except for adding different matrix rosiglitazone nanoparticles. In addition, two fluorescence probes labeled patches (Cou6 and Dir labeled HORN-MN) were similarly prepared by mixing a different probe with the patch matrix.

HORN-MN was photographed by a camera, and Cou6 labeled HORN-MN was observed by a confocal laser scanning microscopy (DIVE, Leica, Wetzlar, Germany) ($E_x = 490$ nm, $E_m = 520$ nm), respectively. A microneedle was manually disrupted to observe the nanoparticles in the microneedle matrix, and the cross-section of the microneedle was photographed by scan electronic microscopy (SEM, Merlin Compact, ZEISS, Oberkochen, Germany).

2.6. Mechanical strength

To evaluate the mechanical strength of the microneedle patch, the patch was placed on a flat metal plate and tested by a texture analyzer (TA. XT Plus, Stable Micro Systems, Godalming, UK). The flat-headed upper stainless-steel probe (1 mm) was set to move toward the patch at a speed of 0.6 mm/min, and the curve of compression force was recorded. The failure force of the microneedle patch was determined as the needle tip began to buckle.

2.7. Transdermal drug release

For measuring transdermal drug release, Cou6 labeled HORN-MN was applied to the skin and then placed on the Franz cell. PBS (pH 7.4, 15 mL) was used as the release medium in the acceptor chamber at 32 °C with magnetic stirring. At the fixed time points, a specimen (200 μ L) was sampled, and the same volume of fresh medium was immediately replaced with the acceptor chamber. At the end of the experiment, the patch was removed from the skin; then the skin was ground into a fine powder to extract the residue Cou6 labeled HORN on the skin. The fluorescent probe (Cou6) in the donor chamber, the acceptor chamber, and the skin were measured by microplate readers (BioTek synergy neo2, Agilent, Santa Clara, CA, USA), respectively. The release rate (%) was calculated by comparing the amount of Cou6 in the acceptor chamber with the added total amount of Cou6.

2.8. Skin status of microneedle puncture

For evaluating the effect of a microneedle patch on the skin, HORN-MN was applied to the in vivo mouse abdomen skin and removed at 5 min for observation by a camera. Besides, Cou6 labeled HORN-MN patch was the same applied to mouse skin and removed at 5 min. The skin status was observed by confocal microscopy. In addition, HORN-MN was applied to the ex vivo mouse abdomen skin and removed at 5 min. The skin was embedded in paraffin, stained with hematoxylin-eosin (HE), and scanned by a digital panoramic scanner to observe the puncture holes.

2.9. Subcutaneous delivery efficiency

For evaluating subcutaneous delivery efficiency, a Dir labeled HORN-MN patch was applied to the in vivo mouse abdomen skin and removed at 5 min for observing drug retention and penetration using the in vivo imaging system (IVIS, PerkinElmer, Waltham, MA, USA). The mice were scanned at 5 min, 1 h, 2 h, 6 h, 12 h, 24 h, 48 h, respectively. The measurement parameters were set as the following: exposure time 4 s, excitation wavelength 745 nm, emission wavelength 800 nm, 1F/stop, 8 binning, and 22.5 cm field of view. Dir fluorescence intensity was quantified by the Living Image Software (v. 4.3.1, PerkinElmer, Waltham, MA, USA). In addition, a Dir labeled HORN with hyaluronic acid gel mixture (HORN-GEL) and RN-MN were included as a control. Besides, the Dir labeled HORN-MN patch was applied to the in vivo mouse abdomen skin and removed at 5 min for evaluating the penetration depth of the nanoparticles into adipose tissue.

The fluorescent tomography (FLIT) mode transillumination imaging was used for scanning at the fixed time points using the IVIS, and subsequently reconstructed in 3D images by Living Image Software. The skin surface was defined as $z = 0$, and three depth layers were selected on the z axis to output the cross-sectional fluorescence signal images, and the fluorescence intensity of each layer was quantified.

In addition, the Cou6 labeled HORN-MN patch was applied to the mouse abdomen skin and removed at 5 min for further evaluating the penetration depth of the nanoparticles into adipose tissue. The mice were sacrificed at 1 h, 24 h, and 48 h for isolating abdominal skin, respectively. The skin sample was embedded in the optimal cutting temperature compound (OCT, mixture of polyethylene glycol and polyvinyl alcohol, Sakura Finetek, Torrance, CA, USA), rapidly cooled in liquid nitrogen for 20 s, and cut into 50 μ m thick sections. The skin cryosections were then observed by fluorescence microscopy.

2.10. Therapeutic efficacy in the treatment of obese mice

For establishing an obese mouse model and dosing, C57BL/6J mice were fed the high fat diet for 92 days. The above mice were randomly divided into four groups and then treated since day 92 with two pieces of RN-MN, HOAN-MN, and HORN-MN at a total dose of 6 mg/kg rosiglitazone every other day 15 times, each time for 5 min, respectively. Body weight and food intake were monitored every other day since day 92.

For assessing treatment efficacy, body fat weight was evaluated in mice at the end of treatment using bone densitometry (Ultrafocus DXA system, Faxitron, Tucson, AZ, USA). The mice were then anaesthetized to collect blood serum. Serum biochemical indicators were tested by a clinical chemistry analyzer to indicate the effect on obesity therapy (BC-430, Mindray Global, Shenzhen, Guangdong, China).

For evaluating preliminary safety, microneedles were applied to the mouse abdomen skin every other day for a total of 15 times. At the end of treatment, the mice were euthanized and the local skin irritation was assessed using HE staining assay.

2.11. Statistical analysis

Statistical analysis was performed by Prism (GraphPad, v. 9.2.0, San Diego, CA, USA). A two-tailed, unpaired Student's t -test was performed to determine the significance between two groups. One-way analysis of variance (ANOVA) followed by Tukey's test was performed to determine the significance for multiple comparison. A value of p less than 0.05 was considered to be statistically significant. Data were represented as mean $\pm$ standard deviation (SD).

3. Results

3.1. Synthesis and identification of nanomaterial

A two-step reaction procedure was used for synthesizing an acid responsive cleaved dimer of hydrazone bond linked oleanolic acid (HOA) (Fig. 1A1 and Fig. 1A2). As a control material, an ester bond linked poly-oleanolic acid trimer (POA) was also synthesized by esterification (without containing a hydrazone bond) (Fig. S1).

HOA was identified by infrared spectroscopy. The characteristic peaks identified were 3532 cm^{-1} (O-H), 2955 cm^{-1} ($-\text{CH}_2-$), and 1693 cm^{-1} (C=O) in OA; 3323 cm^{-1} (O-H), 2932 cm^{-1} , 2876 cm^{-1} ($-\text{CH}_2-$), and 1620 cm^{-1} (C=N) in OH-Hz-OH; and 3336 cm^{-1} (O-H), 2955 cm^{-1} ($-\text{CH}_2-$), 2118 cm^{-1} (C=N-N), and 1687 cm^{-1} (C=O or C=N) in HOA (Fig. 1B).

HOA was further verified by ^1H NMR spectroscopy. The characteristic peaks were fully met by the molecular formula of HOA, which comprised δ : 5.16 (t, $J = 5.2$ Hz, 1H; $-\text{CH} =$) in OA; δ : 6.92 (t, $J = 5.3$ Hz, 1H; $\text{N}=\text{CH}-$), 6.44 (s, 1H; $-\text{NH}-\text{N}$), 4.72 (t, $J = 5.7$ Hz, 1H; $-\text{OH}$), and 3.91 (t, $J = 5.5$ Hz, 2H; $-\text{CH}_2-\text{O}$) in OH-Hz-OH; and δ : 7.90 (m, 1H; $\text{N}=\text{CH}-$), 5.09 (m, 2H; $-\text{CH} =$), and 4.30 (d, $J = 4.8$ Hz, 2H; $-\text{OH}$) in HOA

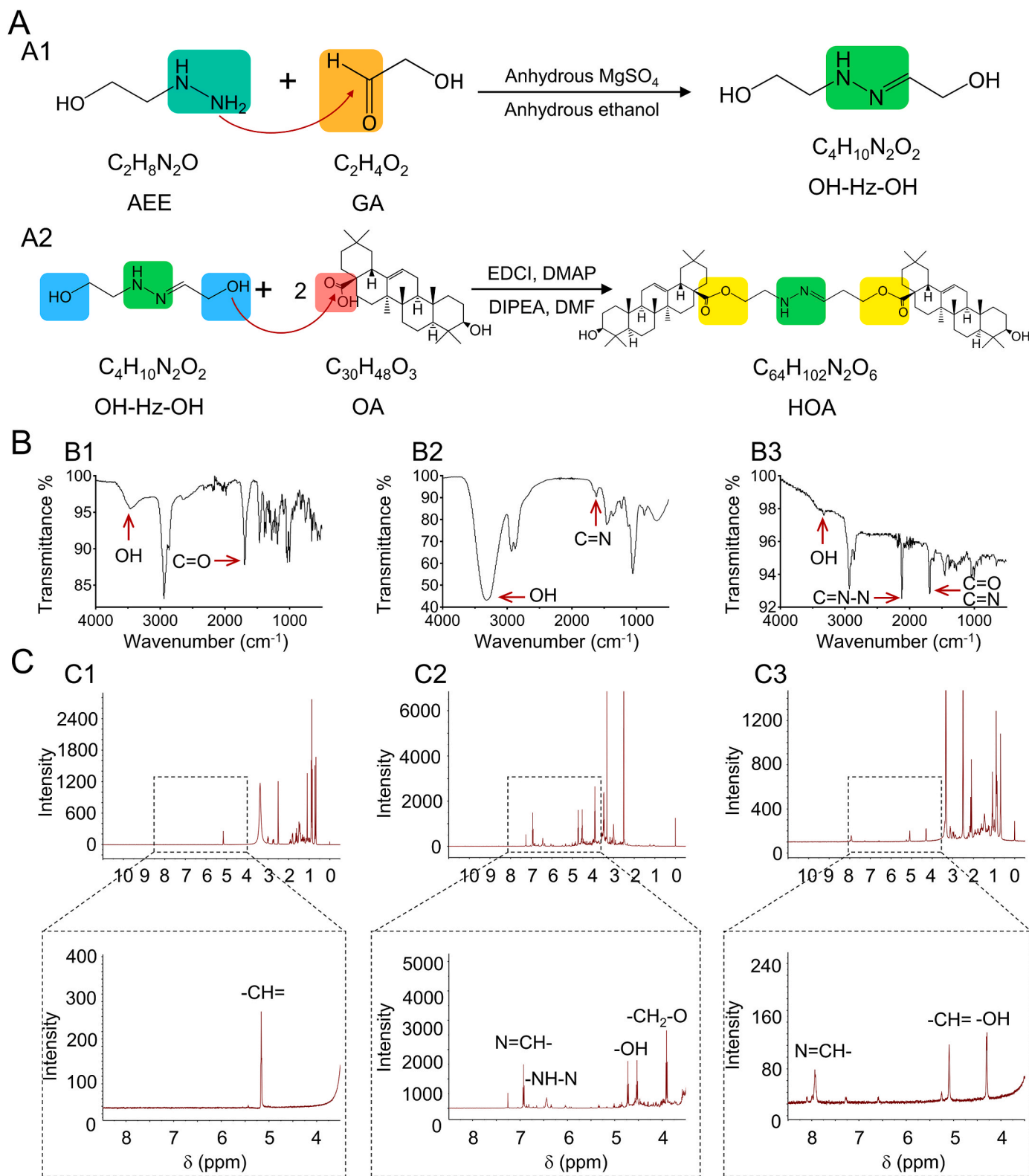


Fig. 1. Synthesis and identification of nanomaterial. **(A) Synthesis route.** (A1) Active intermediate. Glycolaldehyde (GA) was reacted with 2-(2-aminoethylamino)-ethanol (AEE) by nucleophilic addition for producing an active intermediate of 2-(2-hydroxyethyl) hydrazone bond methanol (OH-Hz-OH); (A2) HOA. Hydrazone bond linked oleanolic acid dimer (HOA) was synthesized through the esterification reaction between OH-Hz-OH and oleanolic acid (OA). **(B) Infrared spectrum.** Sample was measured by infrared spectroscopy. (B1) OA characteristic peaks. 3532 cm^{-1} (O-H), 2955 cm^{-1} ($-\text{CH}_2-$), 1693 cm^{-1} (C=O); (B2) OH-Hz-OH characteristic peaks. 3323 cm^{-1} (O-H), 2932 cm^{-1} and 2876 cm^{-1} ($-\text{CH}_2-$), 1620 cm^{-1} (C=N); (B3) HOA characteristic peaks. 3336 cm^{-1} (O-H), 2955 cm^{-1} ($-\text{CH}_2-$), 2118 cm^{-1} (C=N-N), 1687 cm^{-1} (C=O or C=N). **(C) NMR spectrum.** Sample was dissolved in $\text{DMSO}-d_6$ and measured by ^1H NMR spectroscopy. (C1) OA characteristic peaks. δ : 5.16 (t, $J = 5.2\text{ Hz}$, 1H, $-\text{CH} =$); (C2) OH-Hz-OH characteristic peaks. δ : 6.92 (t, $J = 5.3\text{ Hz}$, 1H, $\text{N}=\text{CH}-$), 6.44 (s, 1H, $-\text{NH}-\text{N}$), 4.72 (t, $J = 5.7\text{ Hz}$, 1H, $-\text{OH}$), 3.91 (t, $J = 5.5\text{ Hz}$, 2H, $-\text{CH}_2-\text{O}$); (C3) HOA characteristic peaks. δ : 7.90 (m, 1H, $\text{N}=\text{CH}-$), 5.09 (m, 2H, $-\text{CH} =$), 4.30 (d, $J = 4.8\text{ Hz}$, 2H, $-\text{OH}$). The results indicate a successful synthesis of HOA molecule.

(Fig. 1C and Fig. S2).

POA was identified by infrared spectroscopy, ^1H NMR spectroscopy, and gel permeation chromatography (GPC). The characteristic peaks appeared at 2947 cm^{-1} ($-\text{CH}_2-$) and 1787 cm^{-1} ($\text{C}=\text{O}$) on infrared spectroscopy (Fig. S3) and at δ : 5.16 (t, 1H; $-\text{CH}-$) on ^1H NMR spectroscopy (Fig. S4). GPC calculation indicated that POA had a degree of polymerization of 3 (Fig. S5).

3.2. Engineering of the soluble rosiglitazone nanoparticle microneedle patch

HORN was prepared by linking a hydrazone bond with HOA as a carrier material using a self-assembly nanoprecipitation approach. For the control, an ester bond-linked POA matrix rosiglitazone nanoparticle (ORN), which did not contain an acid cleavable hydrazone, was prepared using POA as the carrier material by the same method (Fig. 2A and Fig. S6). In addition, PLGA matrix rosiglitazone nanoparticle (RN) was prepared using PLGA as the carrier material by single emulsification (Fig. S7).

The physicochemical characteristics of HORN were analyzed by transmission electron microscopy (TEM), dynamic light scattering (DLS), and ultrafiltration. On TEM, all nanoparticles were spherical but had different internal structures (Fig. 2B). Furthermore, rosiglitazone loading increased the size ($102.5 \pm 3.1\text{ nm}$ for HOAN vs $156.9 \pm 9.2\text{ nm}$ for HORN) and zeta potentials ($11.7 \pm 2.6\text{ mV}$ for HOAN vs $18.9 \pm 1.6\text{ mV}$ for HORN) of the nanoparticles. In contrast, RN and ORN had negatively charged zeta potentials ($-59.6 \pm 2.2\text{ mV}$ for RN and $-41.1 \pm 3.3\text{ mV}$ for ORN). The polydispersity index of all nanoparticles was <0.2 , and the encapsulation efficiency of rosiglitazone in all nanoparticles was $>90\%$ (Table S1).

he dispersion stability of HORN was evaluated by the particle size in phosphate buffered saline (PBS) with pH of 7.4 and 5.0. The DLS measurement showed that the particle size of HORN in PBS remained unchanged in pH 7.4 ($141.2 \pm 1.4\text{ nm}$) but evidently increased in pH 5.0, demonstrating a disbanded coarse dispersion system.

Drug release in the different PBS media was determined by high performance liquid chromatography, which evaluated the acid cleavable property. HORN exhibited a higher rosiglitazone release profile in the PBS pH 5.0 medium than in the PBS pH 7.4 medium, indicating a weak acid-responsive release characteristic at 3 h. Conversely, ORN showed a similar rosiglitazone release profile in both PBS media within the same period (Fig. S8).

HORN-MN was engineered by mixing HORN with HA gel and placing the mixture in a microneedle mold, followed by vacuum drying (Fig. 2C). HORN-MN was characterized by photography, confocal laser scanning microscopy (CLSM), and scanning electron microscopy (SEM). On photography, the microneedle array was obviously seen to comprise 15×15 transparent conical needles (Fig. 2D1). Under CLSM, the microneedle was more clearly observed to have a height of $840\text{ }\mu\text{m}$ and a distance of $700\text{ }\mu\text{m}$ between microneedles (Fig. 2D2). After manual fracture, a large number of nanoparticles were observed in a cross section of the microneedle under SEM (Fig. 2D3). After applying to the abdominal skin of mice for 5 min, the HORN-MN patch was removed and observed under CLSM, which showed that the microneedles on the patch already dissolved in the skin interstitial fluid (Fig. 2D4).

3.3. Transdermal delivery of the soluble HORN microneedle patch

The mechanical strength of the microneedles was measured using a texture analyzer for measuring the mechanical yield-force of the microneedles. The mechanical strength of the 20 %, 30 % and 40 % HA microneedle patch was 1.6 N, 2.2 N, and 2.3 N, respectively (Fig. S9). The results of the solubility measurements showed that the 20 % and 30 % HA microneedle patches started to dissolve at 15 s, whereas the 40 % HA microneedle patch started to dissolved at 30 s (Fig. S10). Accordingly, based on mechanical strength and solubility results, 30 % HA was

selected as the microneedle concentration. The yield force per needle was 2.2 N, 1.0 N, 1.3 N, and 0.5 N for HORN-MN, RN-MN, HOAN-MN, and HA-MN, respectively (Fig. 3A); these results indicated that the HORN-MN patch had a suitable mechanical force for application.

The transdermal delivery performance of Cou6-labeled HORN-MN was assessed using a Franz diffusion system, in which freshly isolated mouse skin was placed in the Franz cell system between the donor and acceptor chambers (Fig. 3B1). For comparison, Cou6-labeled nanoparticle hydrogel (HORN-GEL) was included as the control. The results showed that transdermal delivery was rapid for HORN-MN but slow for HORN-GEL (Fig. 3B2, B3 and Fig. S11).

The effect of HORN-MN on mouse skin was evaluated by photography, CLSM, and hematoxylin–eosin (HE) staining. On optical observation, the microneedles were seen to pierce into the stratum corneum with visible fine pinholes. When the patch was removed after 5 min, the microneedles already dissolved in the skin tissue. On further observation of the skin site at 10 min, the pinholes became gradually blurred, the skin was kept intact, and there was no bleeding or other damage (Fig. 3C1). On CLSM, the microneedles left dot fluorescent signals on the skin (Fig. 3C2). On HE staining, the pinholes reached the dermis up to a depth of approximately $450\text{ }\mu\text{m}$ (Fig. 3C3).

The transdermal delivery performance of Dir-labeled HORN-MN was evaluated by an in vivo imaging system (IVIS). Based on the evaluation of the fluorescence signal, both HORN-MN and RN-MN exhibited significant transdermal drug delivery efficiency and sustainable effects for up to 48 h, compared with those of HORN-GEL ($82.0\% \pm 3.0\%$ at 2 h for HORN-MN, $80.0\% \pm 6.3\%$ at 2 h for RN-MN, and $14.0\% \pm 1.1\%$ at 2 h for HORN-GEL) (Fig. 3D1, D2 and Fig. S12). Furthermore, the transdermal delivery depth of Dir-labeled HORN-MN was evaluated by 3D analysis using the IVIS. The fluorescence signal of HORN was observed in the epidermal layer at a depth of 1.5 mm at 1 h, in the dermis layer at a depth of 2.5 mm at 12 h, and in the subcutaneous tissue at a depth of 3.5 mm at 24 h; these indicated gradual diffusion of HORN into the deeper skin layers (Fig. 3D3 and Fig. S13).

The transdermal delivery of Cou6-labeled HORN-MN was further analyzed by fluorescence imaging of the frozen sections of abdominal skin. The fluorescence signal primarily accumulated in the epidermis and dermis layers at 1 h, followed by gradual diffusion into the deeper skin layers and ultimately reaching the subcutaneous adipose tissue at 12 h; the fluorescence signal persisted for up to 24 h (Fig. 3E).

3.4. Regulating of adipose inflammation and transformation of white adipocytes to beige adipocytes

An obese mouse model was established by feeding C57BL/6J mice with high fat diet for 92 days; body weight was monitored to validate the model. On day 92, there was weight gain of $72\% \pm 14\%$ in the obese group and $11\% \pm 1\%$ in the normal control group, indicating successful establishment of the model (Fig. S14). After sacrificing the animals on day 92, the inguinal white adipose tissue (i-WAT) and epididymal white adipose tissue (e-WAT) were collected for RT-qPCR analysis. The results showed that a high fat diet led to evident inflammation, with significantly higher mRNA levels of the inflammatory factors TNF- α , IL-1 β , and IL-6 in both i-WAT and e-WAT than in the normal control (Fig. S15, Table S2).

The patch was applied to the abdominal skin of obese mice 15 times every other day until day 122. The effects of HORN-MN on the genes of adipose inflammation and beige adipocyte transformation in the obese mouse model were assessed by transcriptome sequencing (RNA-seq) analysis. After sacrificing the mice on day 122, i-WAT was collected for RNA-seq analysis. The results showed that compared with RN-MN, HORN-MN caused significant changes in 2168 genes, including upregulation of 1497 genes (69 %) and downregulation of 671 genes (31 %), in the i-WAT (Fig. 4A1). In addition, compared with the blank control, RN-MN caused changes in 1360 genes, including upregulation of 522 genes (38 %) and downregulation of 838 genes (62 %) (Fig. 4A2). The

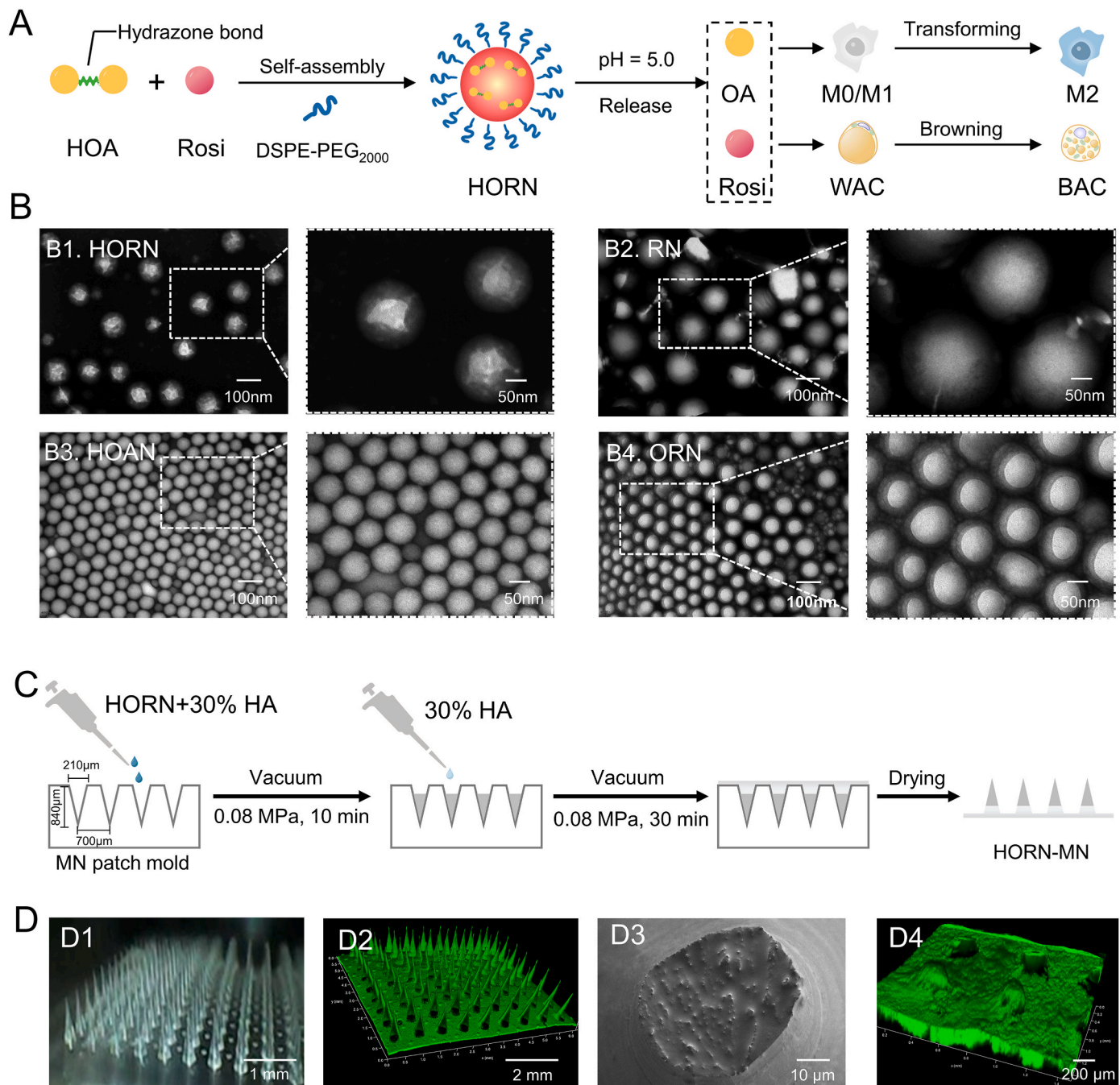
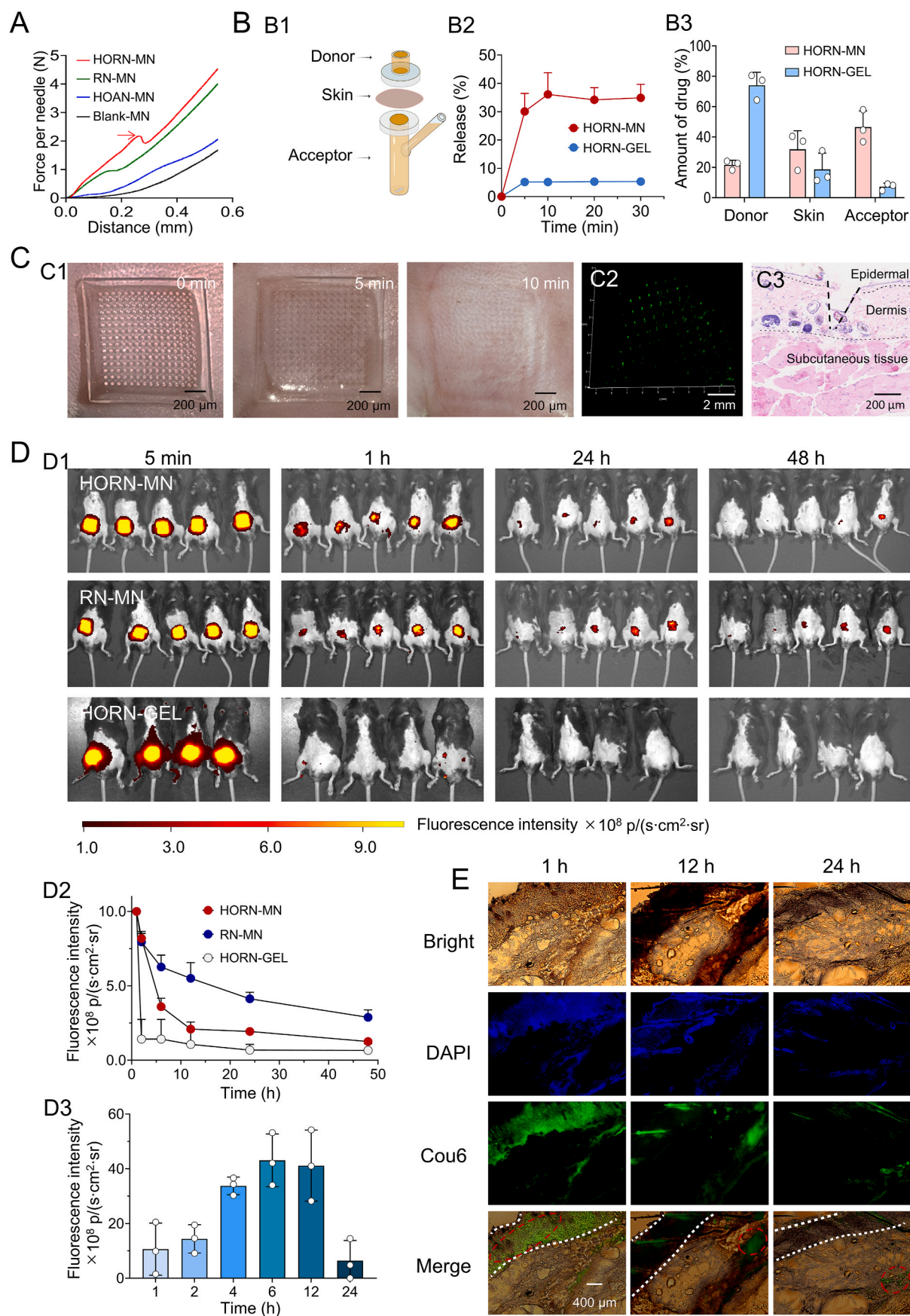


Fig. 2. Engineering of the soluble rosiglitazone nanoparticle microneedle patch. **(A) Design of nanoparticles.** HOA and Rosi were dissolved in the mixture of tetrahydrofuran, acetone, and absolute ethanol, and then mixed with DSPE-PEG₂₀₀₀ to produce HORN through a self-assembly process. Under an acid environment, HORN was disassembled to release Rosi and OA by breaking the hydrazone bond of HOA. OA and Rosi played a role in M0 macrophage remodeling and WAC browning, respectively. **(B) TEM images.** The study was performed by transmission electronic microscopy (TEM). (B1) TEM image of HORN; (B2) TEM image of RN; (B3) TEM image of HOAN; (B4) TEM image of ORN. HORN and HOAN were cage-like nanoparticles, while RN and ORN were spherical nanoparticles. The results indicate a successful preparation of self-assembled rosiglitazone nanoparticle. **(C) Nanoparticle microneedle patch.** HORN in 30 % HA was added to the microneedle patch mold, followed by 0.08 MPa vacuum drying for 10 min to fabricate the tips of the microneedle. For forming the backing layer, 30 % HA was added to the mold, followed by 0.08 MPa vacuum drying for 30 min, and then desiccated at room temperature. The height of the microneedle was 840 µm, and the distance between adjacent microneedles was 700 µm. **(D) Microneedle array images.** (D1) Optical image of microneedles. The image was taken by a camera; (D2) Confocal image of microneedles before use. Cou6 labeled HORN-MN was captured by confocal laser scanning microscopy (CLSM). (D3) SEM image of a disrupted microneedle. A disrupted microneedle was photographed by scan electronic microscopy (SEM). (D4) Confocal image of microneedles after use. After administration to mice for 5 min, the HORN microneedle patch was isolated and observed by CLSM. The results indicate a successful preparation of soluble rosiglitazone nanoparticle microneedle patch in which microneedles exhibit a conical shape with a sharp tip.



(caption on next page)

Fig. 3. Transdermal delivery of the soluble HORN microneedle patch. **(A) Mechanical strength.** Mechanical strength was tested by a texture analyzer to indicate the stiffness of the microneedle ($n = 3$). The red arrow represented the point that the force was suddenly decreased due to the microneedle breakage. **(B) Transdermal drug release.** **(B1) Franz cell.** Mouse skin was equipped onto the Franz cell system between donor and acceptor chambers. Cou6 labeled HORN-MN was inserted in the skin, while Cou6 labeled HORN dissolved in 30 % HA for preparing the gel of HORN (HORN-GEL) served as control was plastered on the surface of skin. **(B2) Drug release profile.** The concentration of Cou6 in acceptor chamber was measured by fluorescence spectrophotometry at the fixed time-points. **(B3) Transdermal drug amount.** At the time point of 24 h in the release experiment, the cumulative release amount of Cou6 in acceptor chamber and the residue Cou6 labeled HORN in donor and skin was measured by fluorescence spectrophotometry, respectively ($n = 3$). **(C) Effect of patch on skin.** (C1) Optical images. HORN-MN was applied to mouse skin for 5 min, and then were removed for observing the skin status for another 5 min, respectively; (C2) Confocal image. Cou6 labeled HORN-MN patch was applied to mouse skin for 5 min and was removed for observing the skin status by CLSM; (C3) Hematoxylin-eosin staining image. HORN-MN patch was applied to mouse skin for 5 min, and then the skin was stained with hematoxylin-eosin (HE) ($n = 3$). **(D) Subcutaneous delivery efficiency.** Dir labeled HORN-MN patch was applied to mice for 5 min, then removed for observing the drug retention and penetration by IVIS; Dir labeled HORN-GEL was used as the control ($n = 4$). (D1) In vivo images. Fluorescence images at the inguinal site of mice were observed at the fixed time points. (D2) Quantification of drug retention. The fluorescence intensities of the above images were used to indicate the drug retention at the inguinal site of mice, and were analyzed by software of living image (v. 4.3.1) on the IVIS system at the fixed time-points ($n = 5$). (D3) Quantification of drug penetration of HORN-MN. The fluorescence signals at 2.5-mm depth in the subcutaneous skin were collected, reconstructed as 3D images by living image software, and then quantified to indicate drug penetration ($n = 3$). **(E) Subcutaneous distribution imaging.** Cou6 labeled HORN-MN patch was applied to mouse skin for 5 min, and then was removed. At 1 h, 12 h, and 24 h, the mouse skin was excised and frozen for imaging the drug distribution in subcutaneous tissue by fluorescence microscopy. DAPI, fluorescent dye 4',6-diamidino-2-phenylindole; blue color, nuclei; green color, Cou6. The results indicate advantageous performances of microneedle patch, including transdermal drug release profile, mechanical strength, and efficient delivery of HORN into subcutaneous tissue. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

results of KEGG enrichment and heat map analyses showed that HORN-MN exerted its effects on adipose inflammation through the NF- κ B pathway and on beige adipocyte transformation through the PPAR γ pathway (Figs. S16A and B and Figs. S17A and B).

The effects of HORN-MN on the mRNA level of the inflammatory factors in subcutaneous i-WAT and e-WAT were assessed by RT-qPCR. The results showed that HORN-MN effectively decreased the mRNA levels of the proinflammatory factors TNF- α , IL-1 β , and IL-6 and remarkably increased the mRNA level of the anti-inflammatory factor IL-10 in the i-WAT (Fig. 4B). In the e-WAT, HORN-MN partially decreased the mRNA levels of TNF- α and IL-1 β and increased the mRNA level of IL-10 (Fig. S18A).

Macrophage polarization was further evaluated by immunofluorescence assay and flow cytometry. On both immunofluorescence assay and flow cytometry, HORN-MN significantly decreased the expression of the M1 macrophage marker (CD11c) and increased the expression of the M2 macrophage marker (CD206) (Fig. 4C and D, and Fig. S18B).

The indicators (UCP1, PPAR γ , PGC-1 α , Cidea, and Adipo) of the effect on beige adipocyte transformation were evaluated by RT-qPCR and Western blot assays. The results in the i-WAT showed that compared with RN-MN, HORN-MN evidently upregulated the mRNA levels and increased the protein expressions of UCP1, PPAR γ , PGC-1 α , Cidea, and Adipo (Fig. 4E, F and Fig. S19). The results in the e-WAT showed that compared with RN-MN, HORN-MN upregulated the mRNA level and increased the expression of UCP1 but had minimal effects on the other indicators (Figs. S18C, D, E and Table S2).

To understand the relationship between inflammation and alteration of adipocyte inflammation, the correlation of the protein indicators of inflammation and beige adipocyte transformation were evaluated by Western blot assay, including phosphorylated extracellular regulated protein kinase 1 and 2 (P-ERK1/2), and phosphorylated cAMP response element binding protein (P-CREB) with beige adipocyte transformation. The results showed that HORN-MN significantly increased the expressions of P-ERK1/2 and P-CREB in both adipose tissues (i-WAT and e-WAT). This indicated that inhibition of the inflammatory signaling pathway and activation of the PPAR γ pathway synergistically promoted adipocyte transformation, as described below (Fig. 4F, Fig. S19 and Fig. S18E).

To reveal the regulation mechanism of ameliorating the inflammatory state and enhancing white adipocytes browning, we established an inflammatory white adipocyte (If-WAC) model by coculturing RAW 264.7 macrophages with mature adipocytes and conducted a comprehensive analysis of cellular uptake, macrophage polarization phenotype, white adipocyte browning markers and associated proteins (Fig. S20). Based on both in vivo and in vitro studies, the process of HORN-MN action can be summarized into three main aspects: (1) Soluble rosiglitazone nanoparticle microneedle patch (HORN-MN) was pierced the

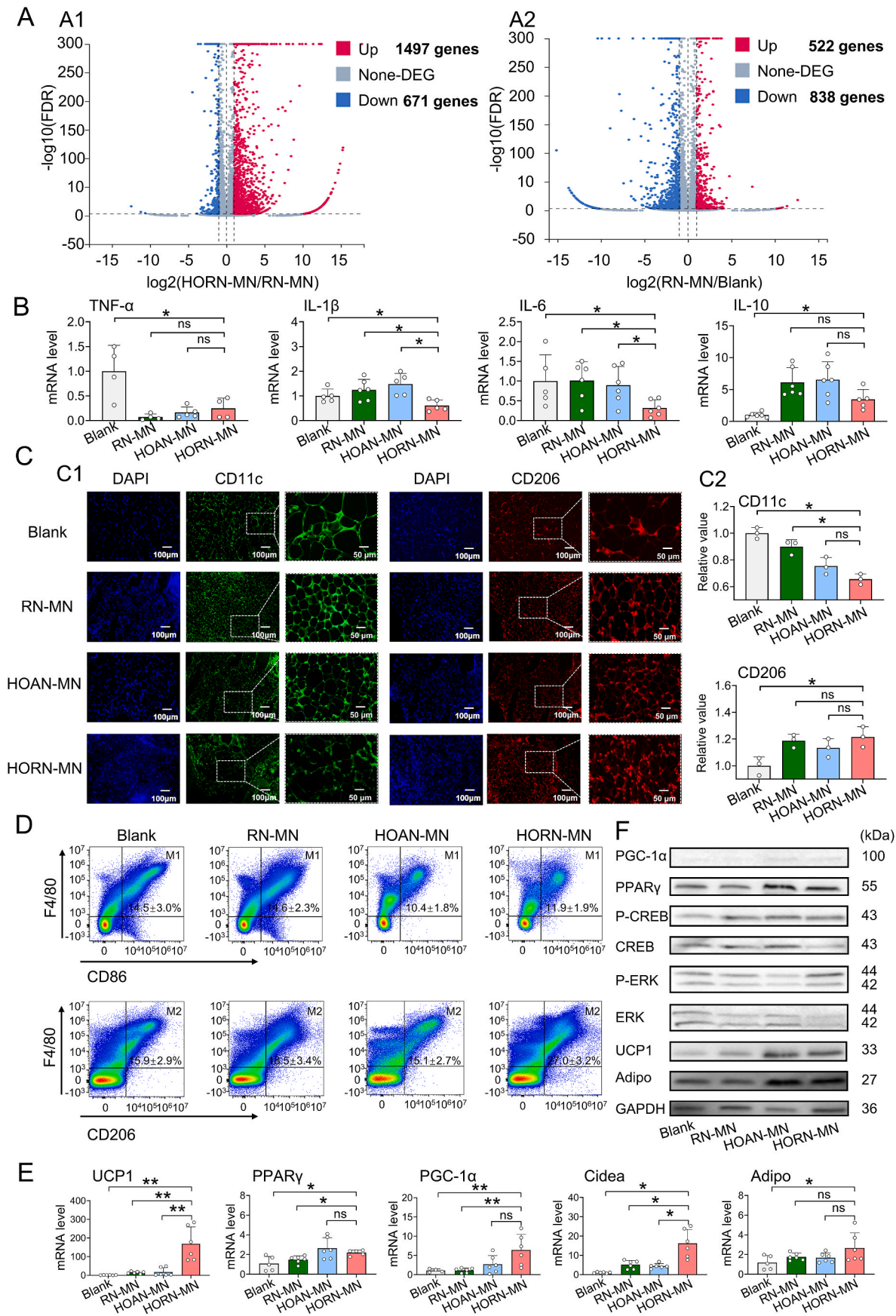
stratum corneum, rapidly dissolved in the tissue fluid of the dermis, followed by releasing nanoparticles; The nanoparticles were endocytosed by macrophages and adipocytes, respectively (Fig. S21). (2) Oleoanolic acid dimer nanoparticle was degraded into oleoanolic acid, which further inhibited the NF- κ B pathway, resulting in the transformation of macrophages from M1 and/or M0 type to M2 type, inhibition of pro-inflammatory cytokines (TNF- α) secretion (Figs. S22 and S23), and secretion enhancement of anti-inflammatory factor (IL-10), thereby improving the microenvironmental inflammation of adipose tissue. Moreover, oleoanolic acid also led to the phosphorylation of ERK1/2 (P-ERK1/2) and CREB (P-CREB), activation of the PPAR signaling pathway, and up-regulation of PGC-1 α and UCP1, thus resulting in a decreased mitochondrial membrane potential, a reduced production of ATP, and the transformation of white adipocytes into beige adipocytes (Fig. S24). (3) Rosiglitazone activated the PPAR signaling pathway, promoted the expression of PGC-1 α and UCP1, and synergistically promoted the transformation of white adipocyte into beige adipocyte (Fig. S25, Fig. 5), resulting in the reduce of TG and further downregulated the expression of IL-1 β , IL-6. The feedback loop between improving the inflammatory environment associated with obesity and promoting the browning of white adipose tissue resulted in a remarkable efficacy of HORN-MN in treating obese mice.

3.5. Treatment efficacy of the soluble HORN microneedle patch in obese mice

Therapeutic efficacy was evaluated by monitoring the body weight of the obese C57BL/6J mice after treatment with various formulations (RN-MN, HOAN-MN, and HORN-MN) (Fig. 6A). The results showed that HORN-MN significantly reduced the body weight by 17 % $\pm$ 3 % after HORN-MN, compared with the effects of RN-MN (7 % $\pm$ 2 %) and HOAN-MN (8 % $\pm$ 2 %) (Fig. 6B). All treatments did not evidently affect the food intake of mice.

At the end of treatment, the mice were anaesthetized to evaluate body fat weight by bone densitometry and then sacrificed to evaluate the weight of the i-WAT and e-WAT. The results showed that HORN-MN significantly reduced the whole-body fat weight by 48 % $\pm$ 4 % (Fig. 6C), indicating its efficacy in reducing both the i-WAT and e-WAT (Fig. 6D1, Fig. S26).

Moreover, the i-WAT and e-WAT underwent immunostaining for UCP1, which is a major indicator of WAC transformation to beige adipocyte. HORN-MN significantly induced the protein expression of UCP1 and decreased the size of adipocytes in both the i-WAT and e-WAT (Fig. 6D2 and Fig. S18D). Moreover, the blood biochemical indicators in the treated mice were measured by a clinical chemistry analyzer. HORN-MN significantly reduced the blood levels of glucose, total cholesterol (TC), triglyceride (TG), low-density lipoprotein (LDL), and high-density



(caption on next page)

Fig. 4. Regulation of adipose inflammation and transformation of white adipocytes to beige adipocytes. **(A) Alterations of adipose genes.** The nanoparticle microneedle patch was applied to mice every other day for 30 days, and then the inguinal white adipose tissue (i-WAT) was collected for transcriptome sequencing. (A1) Volcano map for differential genes for HORN patch treatment; (A2) Volcano map for differential genes for RN patch treatment. Thresholds: $\log_{10}(\text{FDR}) > 5$; and $|\log_2(\text{HORN-MN/RN-MN})|$ or $|\log_2(\text{RN-MN/Blank})| > 1$. **(B) Regulation of inflammation factors.** The above samples were measured by RT-qPCR to indicate the regulating effect on the adipose inflammation microenvironment ($n = 6$). **(C) Phenotype changes of macrophages.** Adipose tissues obtained from mice at the end of microneedle patch treatment were evaluated by immunofluorescence for indicating phenotype change of macrophages ($n = 3$). (C1) Phenotype markers. Each i-WAT was stained by FITC-labeled antibodies (CD11c and CD206) and observed by fluorescence microscopy to indicate the changes in markers. Blue color, nuclei; green color, CD11c; and red color, CD206. (C2) Semi-quantifications. The above fluorescence images were quantified by ImageJ (v. 1.53c) for evaluating CD11c and CD206, respectively. **(D) Quantification for phenotypes of macrophages.** The above samples were further quantified by flow cytometry to evaluate the percentages of M1 and M2 phenotypes ($n = 6$). **(E) Changes of adipocyte browning genes.** The above samples were measured by RT-qPCR ($n = 6$). **(F) Changes of adipocyte browning proteins.** The above samples were measured by Western blot ($n = 3$). The results indicate that HORN treatment results in significant changes of substantial adipose genes, can regulate the adipose inflammation microenvironment, and promote white adipocyte browning. DEG, differentially expressed genes; FDR, false discovery rate. * $p < 0.05$. ** $p < 0.01$, ns, not significant. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

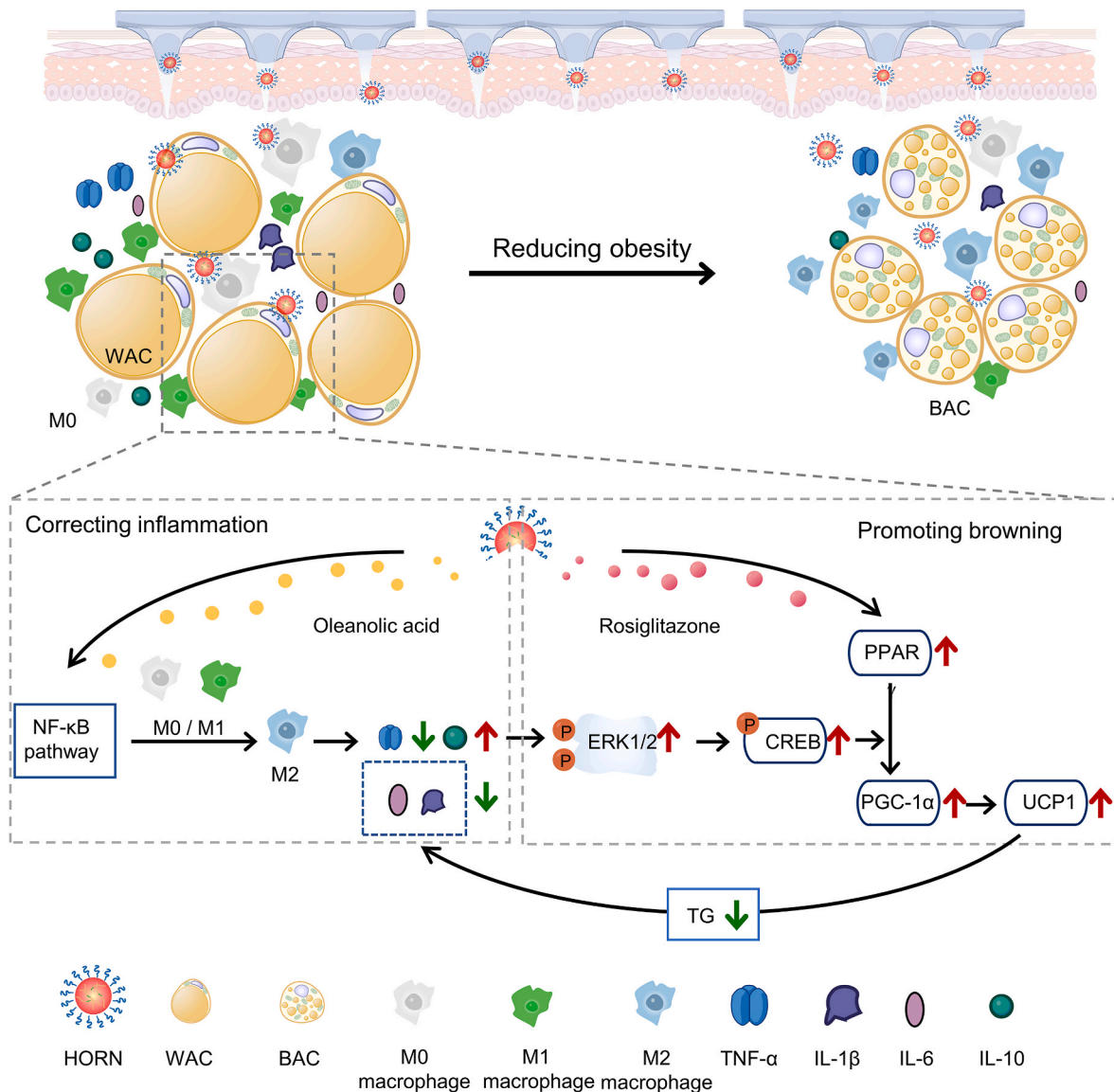


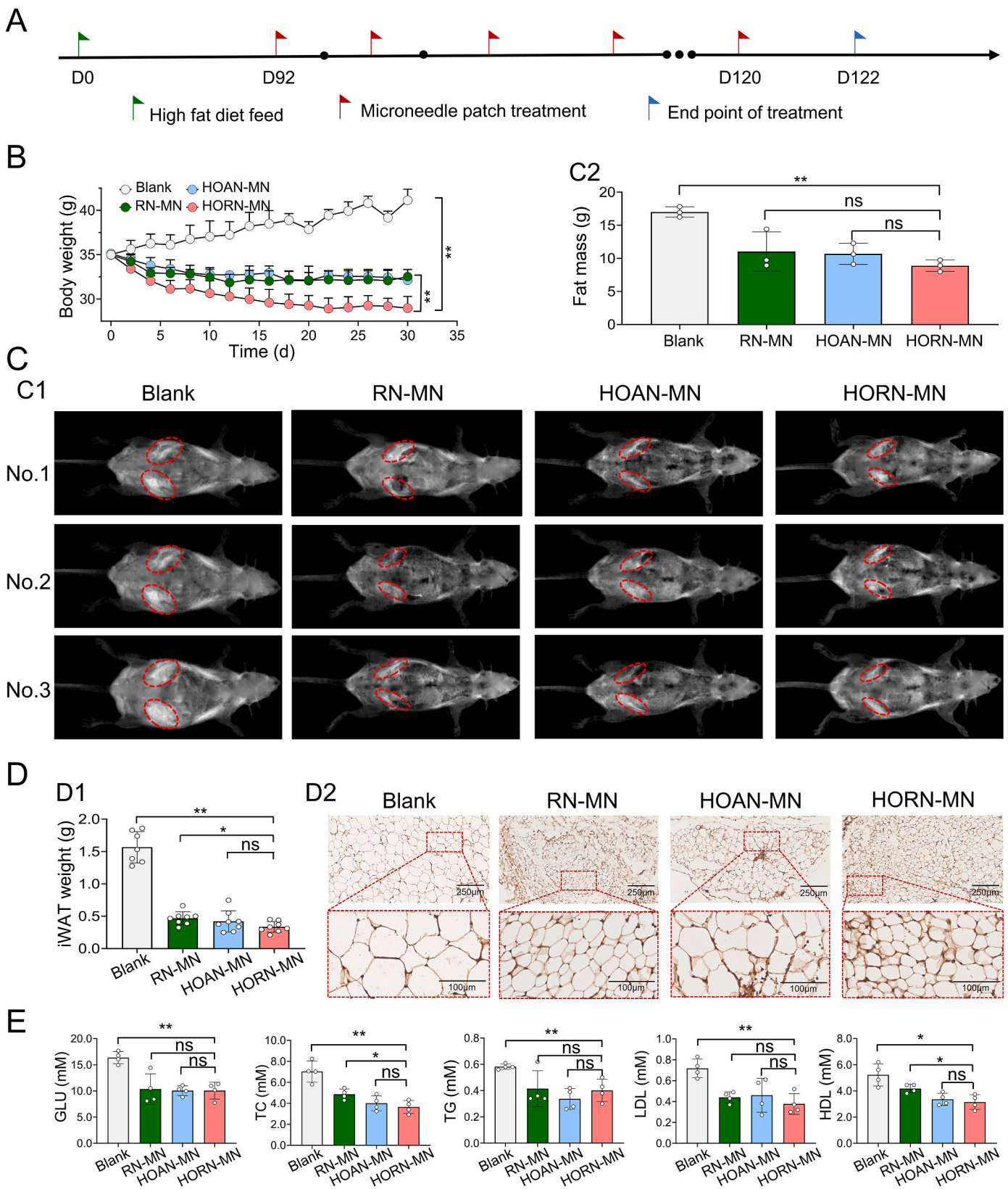
Fig. 5. Mechanism illustration of soluble nanoparticle microneedle patch for the treatment of obesity.

lipoprotein (HDL) (Fig. 6E).

In addition, on HE staining of the skin site with local irritation after HORN-MN treatment, the skin structure remained intact and the structures of the epidermis, dermis, and subcutaneous adipose tissue were clearly seen (Fig. S28).

4. Discussion

Obesity increases the risk of cardiovascular disease, diabetes, sleep apnea, certain cancers, degenerative arthritis, and other diseases [1]. It usually develops from excessive calorie intake, lack of exercise, genetic defects, endocrine abnormality, drug effects, and mental illness [1,19].



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Fig. 6. Treatment efficacy of the soluble HORN microneedle patch in obese mice. **A) Obese model and dosing schedule.** C57BL/6J mice were fed with a high fat diet for 92 days to establish an obese model, randomly divided into four groups ($n = 6$), and then treated since day 92 with two pieces of RN-MN, HOAN-MN, and HORN-MN at a total dose of 6 mg/kg rosiglitazone every other day for 15 times, respectively. **B) Treatment efficacy of obesity.** The therapeutic efficacy of obesity was performed on the above mice by evaluating the body weight every other day since day 92 ($n = 6$). **C) Loss of total body fat weight.** At the end of treatment, the above mice were anaesthetized to evaluate body weight by Ultrafocus DXA ($n = 3$). (C1) Fat map. The body of mice was scanned by Ultrafocus DXA for exhibiting a fat map. Red circle, abdominal fat ($n = 3$). (C2) Quantification of fat weight. Total body fat weight was measured by Ultrafocus DXA to indicate the therapeutic efficacy of obesity. **D) Alteration of i-WAT.** At the end of treatment, the above mice were sacrificed to collect i-WAT. (D1) i-WAT weight. i-WAT was weighed for indicating the loss of subcutaneous adipose tissue ($n = 6$). (D2) i-WAT immunostaining. i-WAT was immune-stained with UCP1 antibody for indicating the white adipocyte browning. ($n = 3$). **E) Serum biochemical index.** At the end of treatment, the above mice were anaesthetized for collecting blood serum. Then serum biochemical indexes were tested by a clinical chemistry analyzer to indicate the effect on obesity therapy ($n = 4$). The results indicate the significant efficacy of the HORN-MN patch in treating obesity. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Development of strategies to reduce obesity by making weight loss easy and increasing patient compliance by self-dosing without effects on health remains challenging [19,20]. In the present study, we developed HORN-MN, which is a soluble rosiglitazone nanoparticle microneedle, patch that can be quickly placed on the abdomen skin and found it to be efficacious, without affecting skin integrity despite multiple administrations.

When designing a microneedle patch, quick dose delivery to prevent prolonged skin contact and local hypersensitivity or irritation should be considered [21]. Given that weight loss often requires a long dosing cycle [22], we reasonably designed a quick paster type of soluble rosiglitazone nanoparticle microneedle patch. First, hyaluronic acid was selected as the microneedle matrix, which had a suitable flexible mechanical strength in a dry state. After processing, the HORN-MN microneedle patch easily pierced and quickly dissolved in the skin before releasing the loaded rosiglitazone. Moreover, the nanoparticle was found to prepare rosiglitazone nanoparticles. The nanoparticle was mixed in the hyaluronic acid matrix to prepare the soluble rosiglitazone nanoparticle microneedle patch. Thirdly, after applying it to the skin, the microneedles of the patch were quickly dissolved in the skin interstitial fluid, and HORN was released. The particle size change of HORN in different PBS solutions (weak acid pH 5.0, and physiological be responsive in a weak acid environment, which is beneficial for drug release in the cellular lysosome. Given that the epidermis and dermis pH is approximately 7.4 [23], this result implied that when HORN was delivered to the epidermal layer of the skin by microneedle, the nanoparticle was stable and intact. After being taken up by macrophages and WAC, HOAN was degraded into OA in lysosomes, then rosiglitazone was released. OA played a role in regulating subcutaneous tissue inflammation by acting on macrophages, whereas rosiglitazone played a role in reducing obesity by acting on WAC.

HOA was synthesized as the initial material to prepare the nanoparticles because of its unique self-assembly capability to prepare the nanoparticle, and the pharmacological effect of its degradation product to treat inflammation [10]. OA and its analog ursolic acid were reported to self-assemble into nanoparticles through hydrogen bonds [24,25]. Considering the encapsulation efficiency and degradation required by the specific purposes of the present study, HOA was newly synthesized as the carrier material, in which oleanolic acid was linked by an active intermediate that provided a hydrazone bond to form oleanolic acid dimer, which was degraded in the weak acid environment of cellular lysosomes [26], thereby, releasing OA and its carried rosiglitazone. Besides, linking HOA with a hydrazone bond demonstrated better flexibility of the self-assembled nanoparticle and higher rosiglitazone encapsulation efficiency. These results were verified by the degradation of HORN in a weak acid environment, as described in detail in the methods sections.

When preparing a soluble rosiglitazone microneedle patch, the molecular weight and concentration of HA might affect the mechanical strength, dissolving rate [13], and dosing time of the microneedles. In this study, we selected HA, which had a molecular weight of 10k Da and a concentration of 30 %, as the matrix material. Under these conditions, HA had a suitable degree of intermolecular stacking and strong hydrogen bond interaction to reach a suitable mechanical strength [27].

The results demonstrated a mechanical strength of 2.2 N/needle, much higher than the required threshold (0.058 N/needle) to penetrate the stratum corneum [28,29]. In addition, HA with a molecular weight of 10k Da has a relatively fast dissolving rate in skin interstitial fluid [13]. This study showed that the patch could be removed after 5 min of transdermal application, while the released HORN was retained in the skin for 24 h and penetrated the subcutaneous adipose tissue, indicating a quick soluble characteristic. Moreover, approximately 30%–40 % of the drug was successfully delivered to the subcutaneous tissue, while another 30%–40 % remained in the skin, indicating that HORN-MN had a high efficacy of transdermal delivery [30]. Accordingly, compared with the regular microneedle patch, which requires 1–6 h of skin application, this HORN-MN patch could be more suitable for the current fast-paced working population [13,30].

In the study, the significant improvement of the inflammatory status of obese adipose tissue after application of the HORN-MN patch can be explained by inhibition of the NF- κ B pathway, macrophage transformation from M1 and/or M0 to M2 type [19,31], inhibition of proinflammatory cytokine secretion, increased secretion of anti-inflammatory factor [32], phosphorylation of ERK1/2 and CREB [33,34], and the subsequent upregulations of PGC-1 α and UCP1 to promote the energy expenditure. Moreover, the patch displayed an effective efficacy in accelerating WAC transformation to beige adipocytes. This can be explained by the action of the released rosiglitazone, which activated the PPAR γ pathway and upregulated PGC-1 α and UCP1 [7,24,35], resulting in dissipation of the proton gradient on the inner mitochondrial membrane, reduced mitochondrial membrane potential, and enhanced energy expenditure [4].

Compared with the previously reported approaches to induce adipocyte browning as treatment for obesity, weight loss was achieved in this study by modulating the proinflammatory microenvironment in adipose tissue. After HORN-MN treatment, the gene levels of inflammatory factors, including TNF- α , IL-1 β , IL-6, were significantly lower than before treatment, and even able to return to normal levels in mice (Fig. S29). By harnessing the immunomodulatory effects of HOA, an impressive weight loss of 17 % was observed; this was significantly higher than reported efficacy of well-known counterparts, such as rosiglitazone alone (10 %) [15,35], rosiglitazone combined with glucose oxidase (10 %) [14], rosiglitazone combined with a photothermal therapy agent (10 %) [36–38], and CL316243 or triiodothyronine (15 %) [39] (Table S3).

5. Conclusions

In summary, we successfully developed HORN-MN, which is a quick paster type for rosiglitazone and OA delivery, and dissolved in the abdomen dermis in 5 min, followed by releasing rosiglitazone nanoparticles. After which, the nanoparticles were endocytosed by macrophage and white adipocyte, and degraded into oleanolic acid in the lysosomes, thereby releasing rosiglitazone. Furthermore, oleanolic acid found it to significantly improve the inflammatory status of obese adipose tissue, promote and enhance WAC transformation to beige adipocytes, and remarkably reduce obesity in mice. This abdominal skin patch can be suitable for quick use by busy working people who consume high-

energy junk foods for efficacious treatment of obesity, without affecting skin integrity despite multiple administrations.

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Ethics approval and consent to participate

All animal experiments were approved by Committee for Experimental Animal Welfare of Biomedical Ethics of Peking University (No. LA2020163).

CRediT authorship contribution statement

Songyue Chen: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Writing – review & editing. **Junbo Wang:** Funding acquisition, Formal analysis, Conceptualization. **Liyu Sun:** Methodology, Formal analysis. **Fangzhi Xia:** Methodology, Formal analysis. **Wenzhe Li:** Methodology, Formal analysis. **Lan Yuan:** Methodology, Formal analysis. **Chang Liu:** Methodology, Data curation. **Peishan Li:** Methodology, Data curation. **Chunjie Bao:** Methodology, Data curation. **Mengjie Wang:** Methodology, Data curation. **Guiling Wang:** Methodology, Data curation. **Jianwei Li:** Data curation, Methodology. **Ying Xie:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Wanliang Lu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biomaterials.2024.122687>.

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